Joint Detection of AI-Generated Images and Post-Processing Alterations in Real-World Scenarios

Giorgia Saolini - 1933076 & Antonio Agostini - 1995653
Computer Vision
Sapienza University of Rome

Outline

1 Problem Statement, Motivation & Research Question

2 Proposed Multi-Task Method

3 Dataset & Preparation Strategy

4 Experimental Setup & Evaluation Protocol

5 Unimodal-Baselines Comparison, Ablations & Results

6 Conclusions & Future Work

Problem Statement & State Of The Art

Problem Statement, Motivation & Research Question

Problem Statement

AI-generated image detectors are evaluated on clean images.

Operations that introduce degradation patterns can hurt forensic detection.

State Of The Art

Existing detectors often focus on authenticity under ideal conditions.

Real-world benchmarks highlight a gap between laboratory accuracy and practical robustness.

Project Goal & Research Question

Problem Statement, Motivation & Research Question

Project Goal

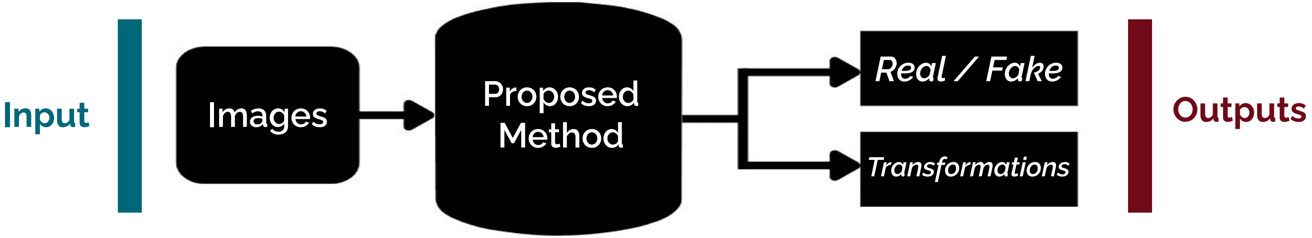
Input:
One single image

Target Outputs:

- Real vs. AI-generated
- Original vs. internet-transmitted vs. re-digitized

Research Question

Can joint learning improve robustness compared to separate unimodal models?

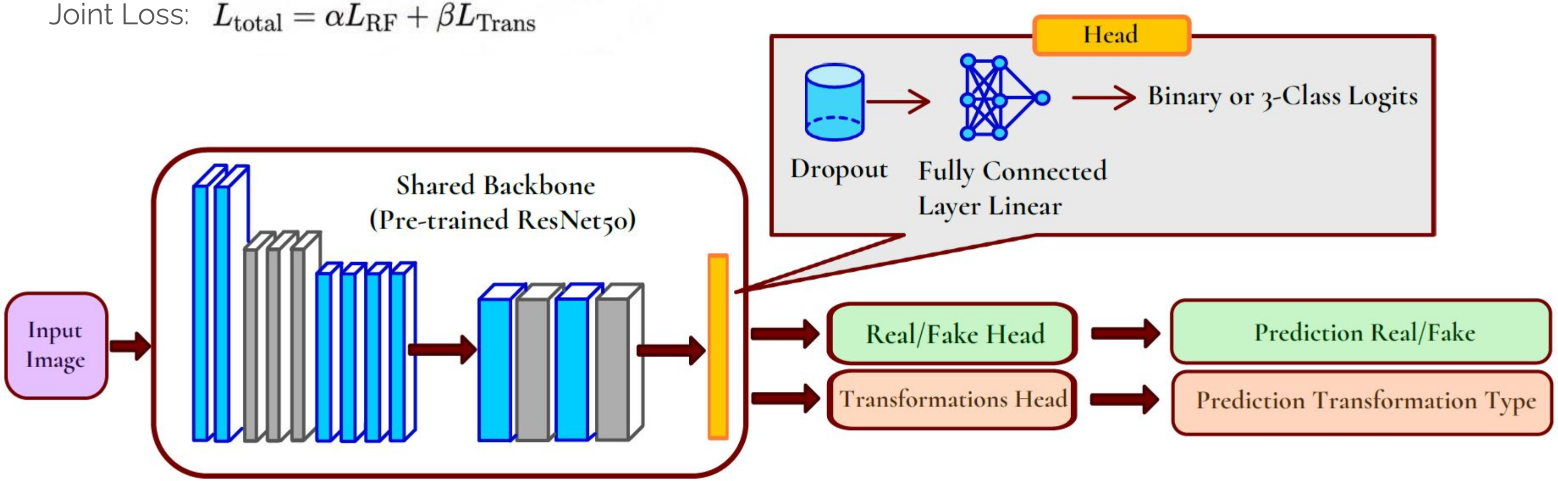


Proposed Method

Proposed Multi-Task Method

- Shared backbone: pre-trained ResNet50
- Final ImageNet classifier removed: *fc = Identity()*
- Two task-specific heads: one for Real/Fake, one for Transformation type
- Transformation classes: original, transfer, redigital
- Joint Loss: $L_{\text{total}} = \alpha L_{\text{RF}} + \beta L_{\text{Trans}}$

IMAGENET



Dataset & Data Preparation Strategy

Dataset & Preparation Strategy

Dataset: RRDataset

- **Two image classes**: Real and AI-generated
- **Three transformation categories**:
  - original
  - internet-transmitted
  - re-digitized
- **Balanced subset selection**: allowed by the project track due to hardware constraints

```
RRDataset
├── original
│   ├── ai
│   │   └── images...
│   └── real
│       └── images...
├── transfer
│   ├── ai
│   │   └── images...
│   └── real
│       └── images...
└── redigital
    ├── ai
    │   └── images...
    └── real
        └── images...
```

Preparation Strategy

- **Built disjoint train / validation / test splits**: from the real benchmark.
- **Balance was preserved across**:
  - Real vs. Fake
  - Original / Transfer / Redigital
- Preferred real benchmark transformations over synthetic on-the-fly simulation.

This approach significantly reduces the domain gap between training and evaluation.

Experimental Setup

Experimental Setup & Evaluation Protocol

Data Configuration

Hyperparameters	Value
Seed	42
Input size	224 x 224
Batch size	32
Optimizer	AdamW
Learning rate phase 1	2e -4
Learning rate phase 2	Heads: 1e -4 ; Backbone: 3e -5
Regularization	Weight decay = 1e -4
Loss weighs	α = 0.9 ; β = 1.4
Label smoothing	0.05
Early stopping patience	3

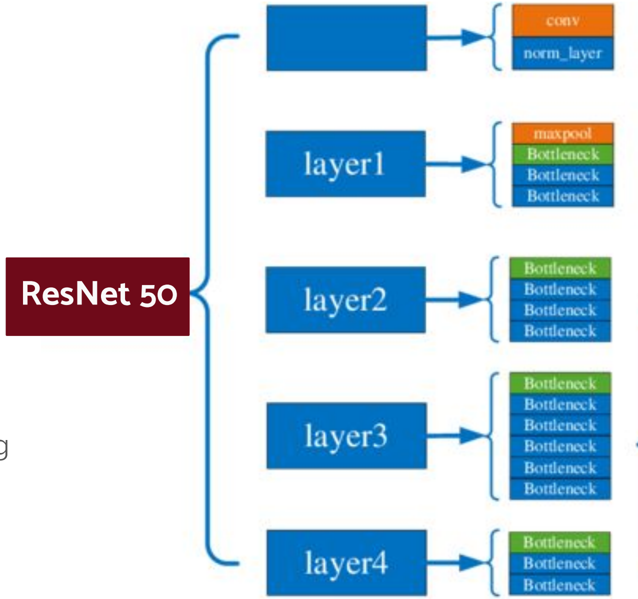
Training & Optimization

Training Protocol

Two-phase: Frozen backbone first, then progressive unfreezing

Optimizer

AdamW with mixed precision, label smoothing (0.05), and early stopping



Training Pipeline Setup

Experimental Setup & Evaluation Protocol

Phase 1 – Head Warm-Up

Backbone fully frozen; only the real/fake and transformation **heads** are trained. This stabilizes the multitask classifier before modifying pretrained features.

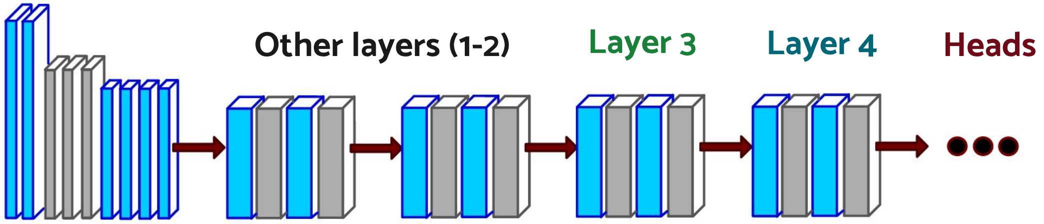
Phase 2.1 – Last-Block FT

`layer4` is unfrozen, while the heads remain trainable. A lower learning rate is used for the backbone and a higher one for heads to preserve knowledge.

Phase 2.2 – Deeper FT

After Phase 2 early epochs, `layer3` is also unfrozen. This enables deeper adaptation of forensic representations without abruptly opening the network.

Why this strategy? Start stable, adapt late features first, then progressively refine deeper forensic representations.



Evaluation Metrics

Experimental Setup & Evaluation Protocol

Metrics	Meaning
Accuracy	Overall percentage of correct predictions
Macro-F1	Balanced performance across all classes
Precision	How many predicted positives are correct
Recall	How many true positives are retrieved
Confusion Matrix	Which classes are confused with each other
Joint Score	Average of Real/Fake and Transformation accuracy

Ablation 1: Loss Weighting

Unimodal-Baselines Comparison, Ablations & Results

Best Configuration (Run 1)

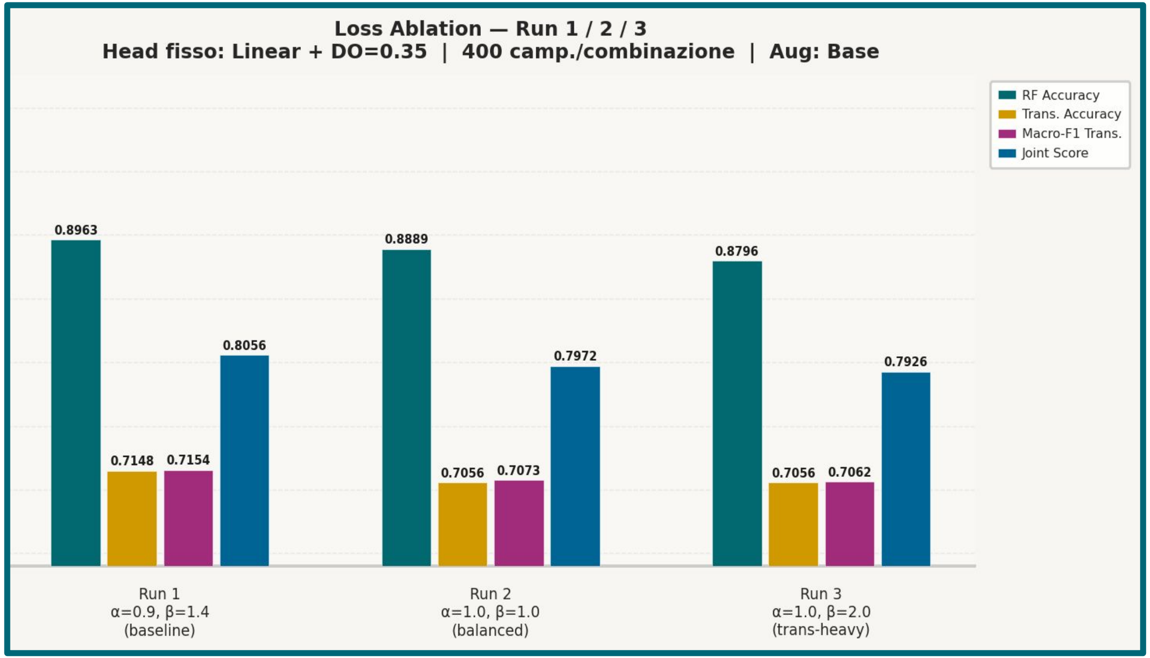
Parameters: α = 0.9, β = 1.4

- RF Accuracy: **0.8963**
- Trans Acc: **0.7148**
- Macro-F1: **0.7154**
- Joint Score: **0.8056**

Key Conclusion

Simply changing the transformation loss weight does not solve the bottleneck.

The auxiliary task requires alternate optimization strategies beyond basic weighting.



Ablation 2: Head Complexity and Dropout

Unimodal-Baselines Comparison, Ablations & Results

Tested Configurations

Ablation: MLP Head & Dropout rates (0.25, 0.50)

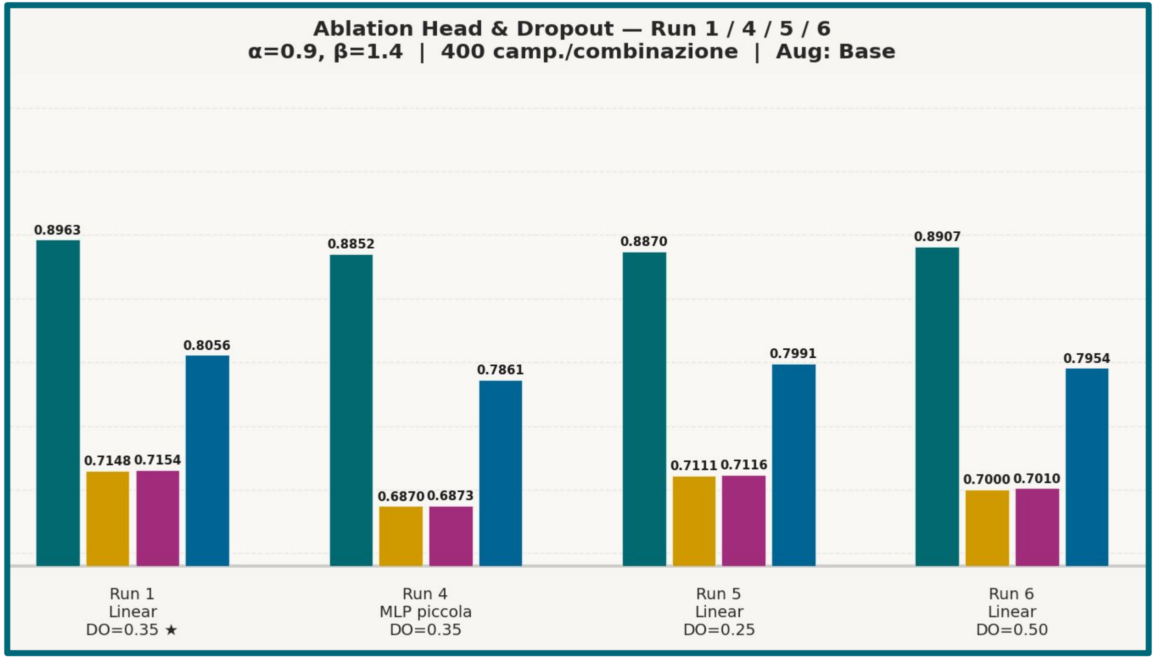
Result: None of these modifications improved results

Underperformed baseline config

Key Conclusion

Best compromise: Linear head + Dropout 0.35

The bottleneck resides in the robustness of the learned features under transformation variability, rather than head capacity limits.



Main Improvement: More Data and Better Augmentation

Unimodal-Baselines Comparison, Ablations & Results

Data-Centric Strategy

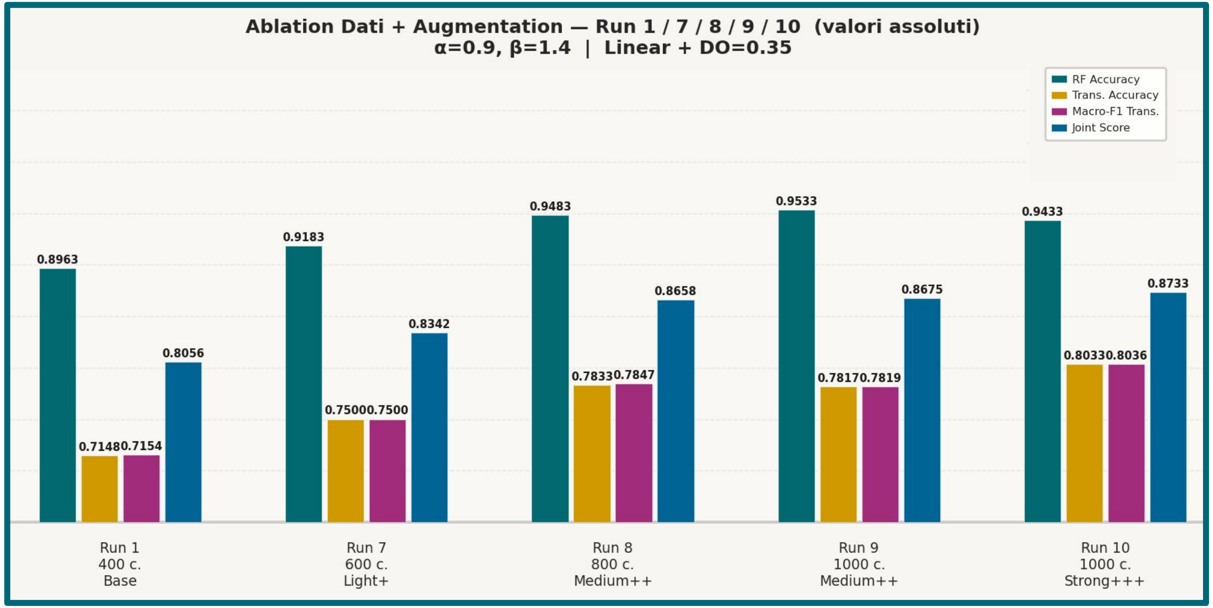
Focus: Feature robustness through data

- Increase samples per combination:
  400 → 600 → 800 → 1000
- Apply progressively richer augmentation

Key Conclusion

Data diversity improves over Head complexity

The model benefited much more from richer data variability than from a more complex classification head.



Best Multi-Task Result

Unimodal-Baselines Comparison, Ablations & Results

Configuration & Augmentation

Best configuration: Run 10

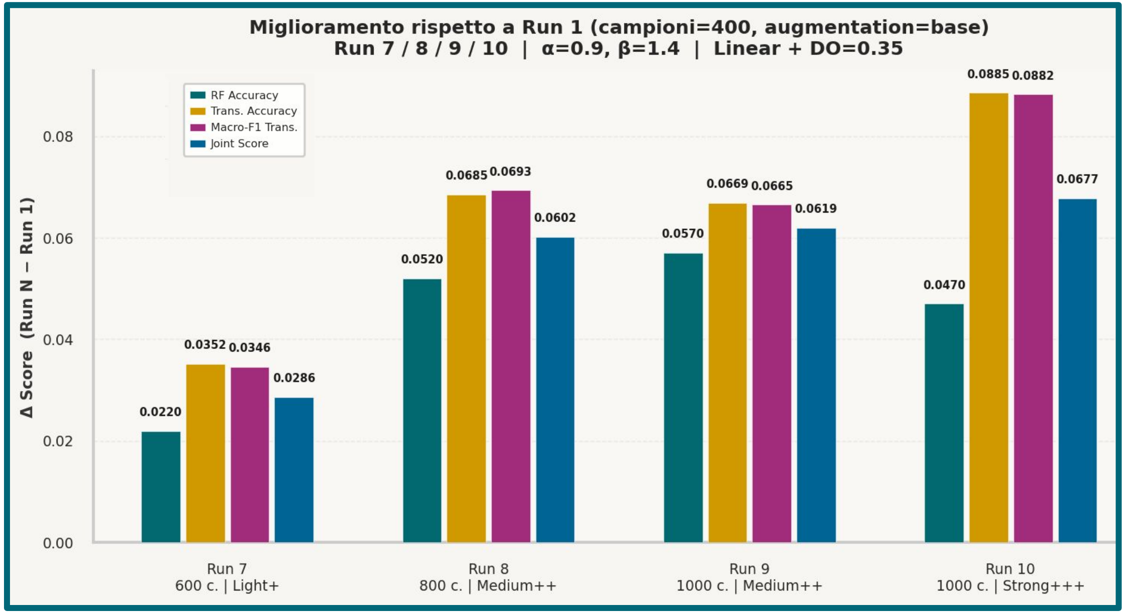
Linear heads, dropout 0.35, alpha 0.9, beta 1.4

1000 samples per combination

Augmentation: RandomResizedCrop, horizontal/vertical flip, small rotation, ColorJitter

Final Test Results

- **RF Accuracy:** 0.9433
- **Transformation Accuracy:** 0.8033
- **Macro-F1 Transformation:** 0.8036
- **Joint Score:** 0.8733



Best Multi-Task Result

Unimodal-Baselines Comparison, Ablations & Results

Configuration & Augmentation

Best configuration: Run 10

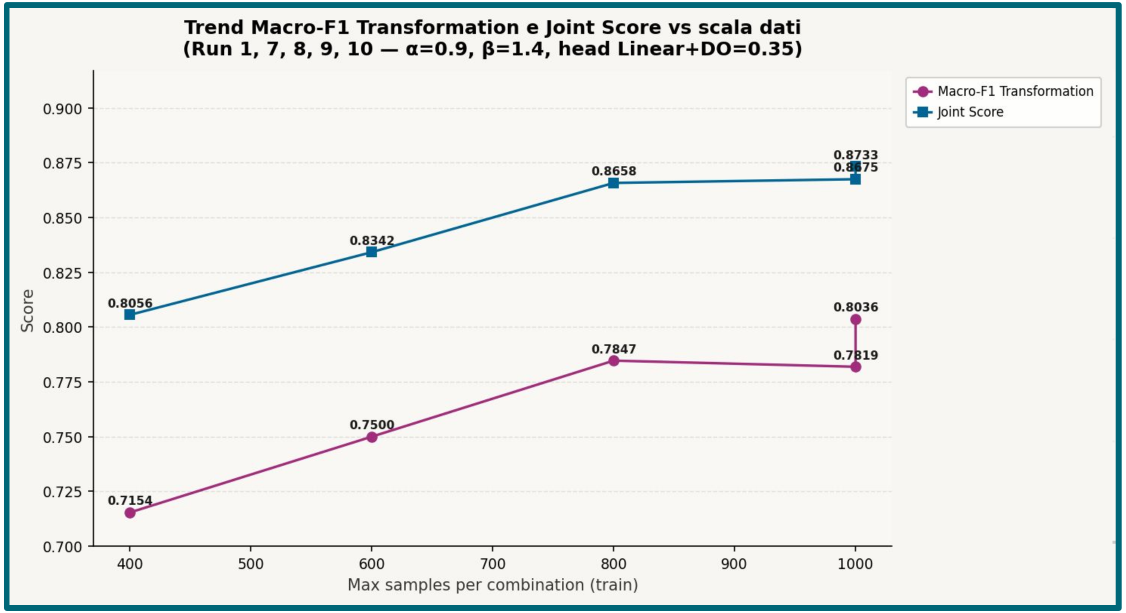
Linear heads, dropout 0.35, alpha 0.9, beta 1.4

1000 samples per combination

Augmentation: RandomResizedCrop, horizontal/vertical flip, small rotation, ColorJitter

Final Test Results

- **RF Accuracy:** 0.9433
- **Transformation Accuracy:** 0.8033
- **Macro-F1 Transformation:** 0.8036
- **Joint Score:** 0.8733



Detailed Evaluation

Unimodal-Baselines Comparison, Ablations & Results

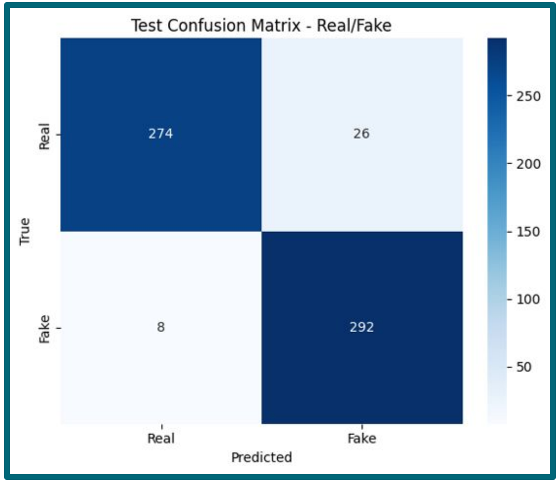
Classification Report

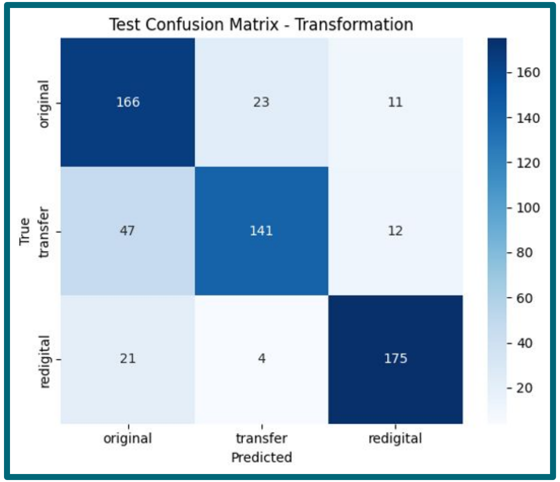
- **Real/Fake:** Accuracy = 0.9433
- **Transformation:** Accuracy = 0.8033

Class Analysis

- **Best Class:** redigital (F1 = 0.88)
- **Hard Class:** original (F1 = 0.76)
- **Transfer Class:** F1 = 0.77

Run	RF Precision	RF Recall	RF F1	RF Macro-F1	RF Accuracy
Run 1	0,92	0,87	0,89	0,90	0,8963
Run 10	0,97	0,91	0,94	0,94	0,9433
Run	**Trans Precision**	**Trans Recall**	**Trans F1**	**Trans Macro-F1**	**Trans Accuracy**
Run 1	0,72	0,71	0,72	0,7154	0,7148
Run 10	0,81	0,80	0,80	0,8036	0,8033





Analyzing the final evaluation in detail, the Real/Fake task proves to be extremely robust, achieving an accuracy exceeding 0.94. Transformation classification is more complex, but still achieves approximately 0.80 in accuracy and macro-F1. The easiest class to detect is the re-digitized one, characterized by a clear degradation pattern, while the original and transfer classes are closer and therefore ambiguous.

Detailed Evaluation

Unimodal-Baselines Comparison, Ablations & Results

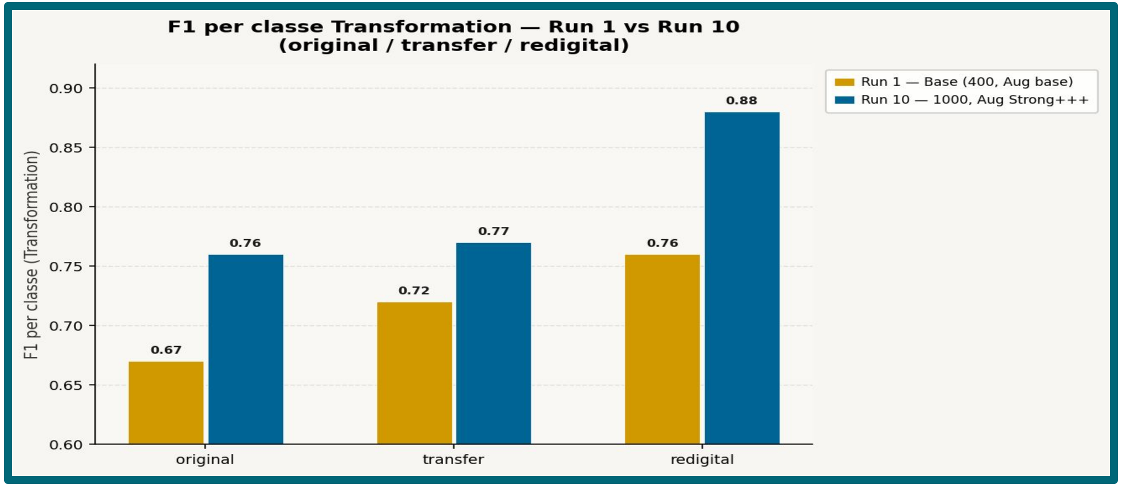
Classification Report

- **Real/Fake:** Accuracy = 0.9433
- **Transformation:** Accuracy = 0.8033

Class Analysis

- **Best Class:** redigital (F1 = 0.88)
- **Hard Class:** original (F1 = 0.76)
- **Transfer Class:** F1 = 0.77

Run	RF Precision	RF Recall	RF F1	RF Macro-F1	RF Accuracy
Run 1	0,92	0,87	0,89	0,90	0,8963
Run 10	0,97	0,91	0,94	0,94	0,9433
Run	Trans Precision	Trans Recall	Trans F1	Trans Macro-F1	Trans Accuracy
Run 1	0,72	0,71	0,72	0,7154	0,7148
Run 10	0,81	0,80	0,80	0,8036	0,8033



Analyzing the final evaluation in detail, the Real/Fake task proves to be extremely robust, achieving an accuracy exceeding 0.94. Transformation classification is more complex, but still achieves approximately 0.80 in accuracy and macro-F1. The easiest class to detect is the re-digitized one, characterized by a clear degradation pattern, while the original and transfer classes are closer and therefore ambiguous.

Multi-Task vs Unimodal Baselines

Unimodal-Baselines Comparison, Ablations & Results

Unimodal Baselines

- **RF-Only:** Accuracy = **0.9517**
- **Transformation:** Accuracy = **0.7933**

Best Multi-Task Model & Trade-off

- **RF Accuracy:** 0.9433 (vs 0.9517 baseline)
- **Transformation Accuracy:** 0.8033 (vs 0.7933 baseline)

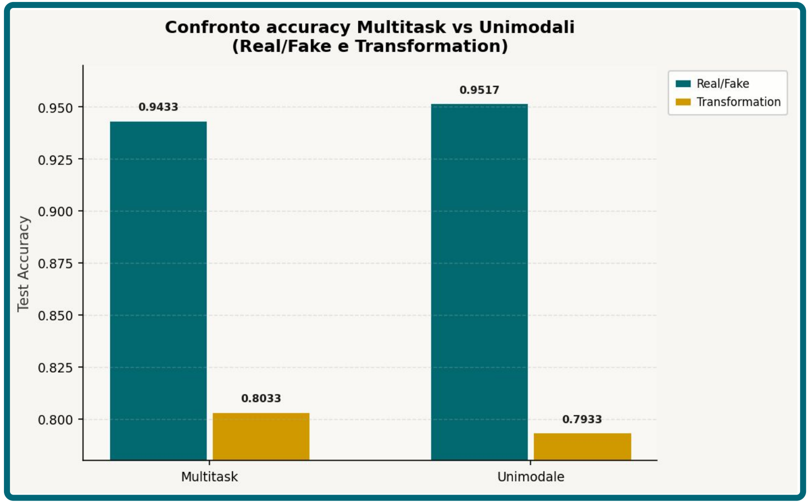
Sacrifices 0.84% on the binary RF task to gain 1% robustness on the harder transformation task.

Task Real/Fake :

Modello	Precision	Recall	F1-Score	Macro-F1	Accuracy
Baseline RF-only	0.95 avg	0.95 avg	0.95 avg	0,9516	0,9516
Multi-task	0.94 avg	0.94 avg	0.94 avg	0,9433	0,9433

Task Transformation :

Modello	Precision	Recall	F1-Score	Macro-F1	Accuracy
Baseline Trans-only	0.80 avg	0.79 avg	0.79 avg	0,7943	0,7933
Multi-task	0.81 avg	0.80 avg	0.80 avg	0,8036	0,8033



The multi-task model achieves comparable performance to specialized baselines on both tasks with a single shared model, demonstrating that the two forensic objectives are not in conflict but rather mutually reinforcing.

Conclusions

Conclusions & Future Work

Key Findings

- A shared-backbone multi-task model is effective for realistic image forensics.
- The main bottleneck was not the head design, but feature robustness under transformation variability.
- More data and targeted augmentation improved generalization significantly.
- Multi-task learning provided the most balanced solution for the full problem.

Project Synthesis

In conclusion, this project shows that a shared-backbone multi-task model is a good solution for realistic AI-image forensics.

The most important lesson is that the central bottleneck was not architectural complexity in the head, but robustness to domain variability in the training data.

By increasing data diversity and using more targeted augmentation, we achieved a clearly stronger and more balanced model, which we believe is the most appropriate result for the full project objective.

Future Work

Conclusions & Future Work

Data Directions

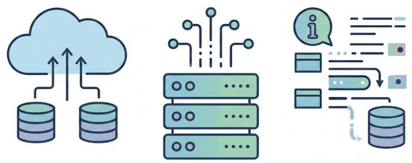
- Stronger semantic or frequency-aware augmentation
- More intelligent sample selection
- Larger-scale evaluation on additional real-world manipulations

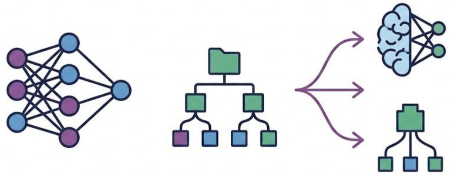
Training Strategy

- Per-transformation curriculum learning
- Transformation-aware training schedules

Model Variants

- Alternative backbone comparison
- Systematic multi-task protocol evaluation





References

Papers

- **Li, Chunxiao, et al.** "Bridging the Gap Between Ideal and Real-world Evaluation..." *IEEE/CVF ICCV*, 2025.
- **Shao, Rui, et al.** "Detecting and grounding multi-modal media manipulation." *IEEE/CVF CVPR*, 2023.

Images

- **ResNet50 Architecture:** https://www.thinkingstack.ai/blog/business-use-cases-11/a-comprehensive-guide-to-resnet50-architecture-and-implementation-56
- **Other Visuals:** All secondary diagrams and schemes were hand drawn.

Slides & Template

- **Sapienza PPT Template:** https://github.com/pietro-nardelli/sapienza-ppt-template

Thank you for the attention!